Dmitry Ivanov

Senior Frontend Engineer

Frontend engineer, 13+ years. Apple/Mobeewave (3.5y) — Apple Pay Later, Samsung POS, Wallet provisioning. Plotly (2y) — WebGL renderers in plotly.js. Brainfinance (3y) — icash.ca, earna.zone design system. Author of audio — a high-level audio workflow library for the web (playback, analysis, non-destructive editing, WASM codecs, 10GB+ files) — and maintainer of the AudioJS ecosystem of DSP packages. Comfortable from GPU shaders and DSP kernels to design tokens.

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github.com/dy

dy (Github)

dmitryiv_ (Twitter)

Work

Senior Frontend Engineer

Brainfinance

Nov 2022 – Oct 2025

Consumer fintech serving 100K+ Canadian users — icash.ca loans, earna.zone savings.

- icash.ca — Core Web Vitals to 95+ PageSpeed via code-splitting, lazy loading, image pipeline
- earna.zone — designed and built the design system from scratch (Shadcn + Tailwind + Storybook)
- Rebuilt the loan contract flow — 40% drop-off reduction, validated by A/B test
- WCAG AA pass across the product — semantic HTML, ARIA, keyboard navigation
- Next.js/Vercel SSR stack, Contentful CMS, Firebase auth

Frontend Engineer (Contract)

Elevenlabs

Oct 2022 – Jan 2023

3-month contract during pre-seed. Ran in parallel with starting at Brainfinance. Company is now valued $1B+.

- Shipped elevenlabs.io — Firebase auth, Stripe payments, 11ty
- Built a multiline waveform player with real-time Web Audio visualization
- Voice generation dashboard UI

Software Engineer

Mobeewave / Apple

Jan 2019 – Oct 2022

Mobile point-of-sale fintech. Acquired by Apple in 2020; stayed through the transition and shipped Apple Pay features to iOS users globally.

- Samsung POS — tap-to-pay web app shipped globally in Samsung Wallet, PCI-DSS compliant
- Apple Pay Later — checkout widget with real-time calculations, A/B tested flows
- Apple Wallet provisioning UI flow and Business Registry portal
- Acquirer portal and admin tools used by 50+ financial institutions
- Spect — internal AOP framework for frontend state and component composition
- Template Instantiation polyfill — reference implementation for the W3C proposal
- E2E test suites in Cypress and Puppeteer

WebGL Engineer

Plotly

Sep 2016 – Sep 2018

WebGL core of plotly.js — the open-source scientific charting library (17K+ stars, used across research and industry).

- WebGL renderers for scatter, line, and waveform traces — interactive at 1B+ points
- Point-clustering algorithm enabling real-time exploration of massive datasets
- Mobile build — touch interactions, responsive layouts
- Contributions to regl and the stack-gl ecosystem (modular WebGL)
- Visual regression harness covering 2000+ chart variants

Web Developer

TTBA Group

Jan 2015 – Aug 2016

Boutique web agency — performance-first client work.

- 10+ client sites shipped at 100/100 PageSpeed
- Wishbox.gift — Preact e-commerce platform with custom API

Frontend Developer

KudaGo

Mar 2012 – Jan 2015

Cultural events platform across 15+ European cities.

- kudago.com desktop + mobile — <20KB JS bundle, 100/100 PageSpeed
- UX/UI and information architecture

Web Developer & Designer

Freelance

Mar 2012 – Mar 2014

Design + frontend for 15+ small/medium business sites — e-commerce, real estate, corporate, analytics dashboards.

Education

Baltic State Technical University "Voenmeh"

Computer Graphics & Intelligent Systems

Sep 2010 – Jun 2012

Master

Baltic State Technical University "Voenmeh"

Information and Computer Science

Sep 2006 – Jun 2010

Bachelor

Saint-Petersburg College of Marine Instrumentation

Automatic Control Systems

Sep 2002 – Jun 2006

Technician

Courses

- Signal Processing

- Control Theory
- Electronics
- Instrumentation

Projects

Wavefont / Linefont

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Data-rendering fonts — draw waveforms and line charts as text. Used in production by plotly.js and others.

audio

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High-level audio workflow library for the web — playback, analysis, non-destructive editing.
`audio('voice.mp3').trim().normalize('podcast').fade(0.3, 0.5).save('clean.mp3')` — chainable, streaming, tree-shakable.

- WASM codecs loaded on demand — no ffmpeg dependency
- Streaming playback begins during decode; page cache handles 10GB+ files
- Non-destructive virtual edits with instant undo
- Audio-first API — works in dB, Hz, LUFS, not byte offsets
- LUFS loudness, spectrum, peak, and descriptor analysis
- Pluggable op architecture; CLI with Unix pipes and tab completion
- Runs in browsers, Node, Deno, Bun

AudioJS ecosystem

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Open-source audio/DSP toolchain for the web — author and maintainer. Modular packages covering decode, encode, I/O, filters, effects, and the full signal path.

- web-audio-api — portable Web Audio API implementation; polyfill compatible with tone.js
- audio-decode / audio-encode — 13+ formats in Node and browser via WASM codecs
- audio-speaker / audio-mic — cross-platform audio I/O
- digital-filter — IIR/FIR/biquad/butterworth, smoothing, adaptive, multirate
- audio-filter — weighting curves (A/C/K), auditory, analog, speech, EQ
- audio-effect — modulation, dynamics, delay, spatial, distortion (phaser, flanger, chorus)
- pcm-convert, periodic-function, audio-buffer-utils, audio-stretch, audio-lena

GLSL-transpiler

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GLSL → JavaScript compiler (stack-gl ecosystem). Lets you run shader code on CPU.

SciJS FFT

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Performant FFT implementation for signal processing.

Sprae

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3KB DOM microhydration library — reactive templating without a virtual DOM.

Watr

Invalid Date – Present

WebAssembly text-format compiler.

Subscript

Invalid Date – Present

Fast expression parser and evaluator.

Color-space

Invalid Date – Present

Color space conversions library — comprehensive, dependency-free.

Skills

Frontend Core

typescript react next.js preact vue signals web components SSR

Graphics, Audio & DSP

webgl glsl regl three.js canvas web audio api waveform rendering signal processing fft wasm

Performance & Testing

core web vitals lighthouse profiling playwright cypress jest puppeteer bundle optimization

Design Systems & UI

shadcn tailwind storybook material-ui figma wcag aria responsive design

Build & Infra

webpack rollup esbuild vercel github actions

Backend

node.js python graphql firebase postgres

Languages

English

fluent

Russian

native

French

basic

References